



[WWW'25 Short Paper] Decision notification for your submission 2801

From WWW25 - Short paper track <WWW25-shortpaper@easychair.org>

Date Mon 20/01/2025 19:00

To Kieu Hai Dang (CECS) <dang.kh@vinuni.edu.vn>

Dear Dang Kieu,

We regret to inform you that your paper, "Keyword-driven Retrieval-Augmented Large Language Models for Cold-start User Recommendations", was not accepted for presentation in the Short Paper track at TheWebConference 2025. The selection process was highly competitive this year, and we sincerely appreciate the effort and dedication you put into your submission.

Workshop Fast-track Opportunities

While your paper was not accepted for the short paper track, we encourage you to consider submitting it to one of the 26 workshops taking place as part of TheWebConference 2025.

Several of these workshops are participating in a fast-track submission process, allowing you to share your existing reviews with the workshop organizers. This enables them to assess your paper's suitability for acceptance without requiring an additional review.

If you are interested in this opportunity, please visit <https://www2025.thewebconf.org/workshop-fast-track> for a list of participating workshops. The deadline for submission is 11:59 PM, January 26, 2025 (AoE).

We sincerely thank you for your submission and hope to see your work featured in other venues at the conference.

Best regards,

Craig, Yiqun, Yadan

SUBMISSION: 2801

TITLE: Keyword-driven Retrieval-Augmented Large Language Models for Cold-start User Recommendations

----- METAREVIEW -----

RECOMMENDATION: reject

TEXT:

This paper proposes a framework to address the cold-start user recommendation problem. The proposed approach uses keywords provided by users for candidate retrieval and uses LLM to re-rank the candidates. The reviewers overall like the proposed approach but raise a number of problems that should be addressed: (1) not enough details on how to select the keywords and how their quality influences the ranking; (2) not enough details on how the few shot prompts are constructed and how the ordering of keywords and candidates influence the performance.

----- REVIEW 1 -----

SUBMISSION: 2801

TITLE: Keyword-driven Retrieval-Augmented Large Language Models for Cold-start User Recommendations

AUTHORS: Dang Kieu, Minh Duc Nguyen, Thanh-Son Nguyen and Duy Dung Le

----- Review. -----

The paper introduces KALM4Rec (Keyword-driven Retrieval-Augmented Large Language Models for Cold-start User Recommendations), a novel framework specifically designed to tackle the challenge of cold-start user recommendations. It requires only a few input keywords from users, effectively addressing the issue of new users lacking historical data. KALM4Rec employs a two-step strategy: first, keyword-driven retrieval models are used to identify potential candidate items; then, large language models (LLMs) combined with various prompting strategies (such as zero-shot and few-shot techniques) re-rank these candidates. This approach not only overcomes the limitations of LLMs in processing extensive tokens but also reduces the risk of generating misleading information. However, The effectiveness of KALM4Rec largely depends on the quality of keywords provided by users. If users cannot accurately describe their needs or preferences, it may lead to suboptimal recommendation results. The refore, guiding users to better express their intentions is a significant challenge. In addition, although the method has been validated on specific datasets(Yelp.com dataset and TripAdvisor dataset, lack online A/B tests), its generalizability has not been fully tested. Different domains (e.g., e-commerce, news information) might present unique challenges, necessitating further research to ensure the wide applicability of KALM4Rec.

----- Overall evaluation -----

SCORE: 5 (Marginally below acceptance threshold)

----- Reviewer's confidence -----

SCORE: 3 (I am fairly confident that the evaluation is correct)

----- REVIEW 2 -----

SUBMISSION: 2801

TITLE: Keyword-driven Retrieval-Augmented Large Language Models for Cold-start User Recommendations

AUTHORS: Dang Kieu, Minh Duc Nguyen, Thanh-Son Nguyen and Duy Dung Le

----- Review. -----

This work addresses cold-start recommendation by using keyword extraction from reviews followed by LLM-based candidate re-ranking. The experimental results demonstrate that this approach outperforms direct review-based recommendations.

The paper can be further strengthened by considering the following aspects:

1. The mechanism behind LLM's effective re-ranking performance remains unexplored, particularly regarding whether the model's pre-trained knowledge of items influences its ranking decisions.
2. The hypothesis that keywords provide cleaner signals than full reviews requires empirical validation through comparative analysis.
3. The methodology for keyword ordering needs explicit description, as both keyword order and candidate order impact system performance, as mentioned in the discussion.

----- Overall evaluation -----

SCORE: 7 (Good paper, accept)

----- Reviewer's confidence -----

SCORE: 2 (I am willing to defend my evaluation, but it is quite likely that I did not understand central parts of the paper.)

----- REVIEW 3 -----

SUBMISSION: 2801

TITLE: Keyword-driven Retrieval-Augmented Large Language Models for Cold-start User Recommendations

AUTHORS: Dang Kieu, Minh Duc Nguyen, Thanh-Son Nguyen and Duy Dung Le

----- Review. -----

This paper introduces a novel framework, KALM4Rec, which addresses the cold-start problem in recommender systems. The proposed method improves cold-start recommendations by leveraging keyword-based initial candidate retrieval and uses Large Language Models (LLMs) for

re-ranking.

Strength:

- The paper presents the method clearly.
- the figures help a lot in understanding the proposed method.
- the proposed method is novel.
- in the LLM reranking stage, various LLM models are experimented with different prompting strategies to guide the LLM in re-ranking the items.

weakness:

- While the paper introduces the idea of a heterogeneous graph with three types of nodes (users, items, and keywords), it is unclear how the graph is constructed and how to define the edges. Specifically, what thresholds or criteria are used to decide when a user's keywords are sufficiently related to an item to form an edge? Is there any weighting applied to edges between keywords and items, and if so, how is this weighting determined?
- The paper does not fully elaborate on how the aggregation of information between nodes (keyword and item) affects the final retrieval results. Why choose the simple-weighted sum function rather than the graph convolutions or other methods here? How does the aggregated feature vector affect the ranking?
- Not clear how the prompts, especially the few-shot prompts are structured.
- The results presented in the experiments do not include statistical significance tests.

Minor issue: The dataset names are not consistent across results tables;

----- Overall evaluation -----

SCORE: 6 (Marginally above acceptance threshold)

----- Reviewer's confidence -----

SCORE: 3 (I am fairly confident that the evaluation is correct)